

EventSphere

Product Requirements Document (PRD)

One digital home for every event.

Version 1.0.0-draft

Status: Draft for review

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Field	Value
Product	EventSphere
Version	1.0.0-draft
Status	Draft for review
Last updated	2026-07-25
Owner	Product
Audience	Engineering, Design, AI/ML, Security, Go-to-Market

Tagline

We don't replace your camera. We don't replace your photographer. We don't replace Instagram.
We give every event one digital home.

Document map

#	Section	File
1	Product Overview	Vision, problem, audience, evidence, hypothesis, success metrics
2	User Roles & Permissions	Roles, permission matrix, access models
3	Feature Catalog	Every feature by use case and platform capability
4	Screens & Navigation	Complete screen inventory + navigation map
5	User Flows	End-to-end flows for all primary actors
6	State Diagrams	Event, media, RSVP, job, and account states
7	Business Rules	Domain invariants and policy
8	Validation, Errors & Edge Cases	Validation, error copy, exceptions
9	API Contracts	REST endpoints, payloads, status codes
10	Database Schema	Entities, relationships, indexes
11	Notifications	Channels, triggers, templates

#	Section	File
12	File Handling	Upload, processing, retention, delivery
13	Integrations	Identity, storage, AI, calendar, social
14	Security Requirements	AuthN/Z, privacy, biometrics, compliance
15	Performance Requirements	Latency, throughput, scale
16	Acceptance Criteria	Testable must-pass criteria
17	Wireframes	Structural mockups for every key screen
18	Glossary	Canonical terminology
19	Assumptions & Open Questions	Decisions still needed

Strategic framing (pitch guidance)

Do **not** present Friends / Celebrations / Professional as pricing tiers in go-to-market narrative. Frame them as **use cases** that grow with event size and complexity. Monetization may map underneath later; the product story is use-case-first.

Assumed platforms (v1)

Platform	Priority	Notes
Progressive Web App (mobile + desktop)	P0	Primary surface for guests via QR
iOS native app	P1	Full organizer + guest experience
Android native app	P1	Full organizer + guest experience
Live photo wall (browser kiosk)	P0 for Celebrations+	Large-display mode
Admin console (web)	P1	Platform operations

Out of scope for v1

- Full social network / public feed outside events
- Marketplace for photographers
- Hardware (physical cameras, NFC badges) beyond QR
- Live video streaming of the event itself
- Ticket sales / payment gateway for event tickets (RSVP only in v1 Celebrations)

How to use this PRD

1. Product & design: start with Overview → Features → Screens → Wireframes → Flows.
2. Engineering: Roles → Schema → API → File Handling → Security → Performance → Acceptance.
3. Legal / privacy: Security → Business Rules (face data) → File retention.
4. QA: Acceptance Criteria → Validation/Errors → Edge Cases → Flows.

1. Product Overview

1.1 Vision

EventSphere is the **digital home for every event** — a single AI-powered platform where invitations, photos, videos, certificates, highlight reels, and personalized memories live together.

People already document everything. The problem is not capture. The problem is **everything that happens after the photo is taken**: collection, organization, discovery, and distribution.

1.2 Problem statement

Every event ends the same way: memories are scattered across phones, messaging apps, cloud drives, and social media.

Small events (friends)

Five friends go to dinner or on a trip. Everyone takes photos. Afterward:

- "Can you AirDrop me those?"
- "Who has the group photo?"
- "Can you send me the videos?"

One person spends hours collecting and sending. Everyone else downloads hundreds of pictures to keep a handful. The best memories still get lost.

Mid-size events (celebrations)

Weddings and parties increasingly use disposable cameras for candid moments — then someone must collect cameras, develop film, organize hundreds of photos, and redistribute them. Guests already carry high-quality cameras in their pockets.

Large events (professional)

At a company dinner, 5,000+ professional photos are dumped into a Google Drive. An attendee who appears in two photos spends ages scrolling to find them.

Core problem: Taking photos is solved. Collecting, organizing, finding, and delivering the right memories to the right people is not.

1.3 Product hypothesis

People do not want another photo gallery.

They want **one platform that manages every digital memory of an event**, where:

- Guests don't chase photos
- Organizers don't chase guests
- Everyone receives the memories that matter to them **automatically**

1.4 Target audience (growth path)

Stage	Audience	Event types
1	Friend groups	Trips, dinners, small gatherings
2	Hosts & celebrants	Birthdays, weddings, university events, private celebrations
3	Corporate & event teams	Corporate events, conferences, festivals
4	Content professionals	Photographers, media teams, influencers, brands, social media managers

1.5 Evidence & market context

- People already document vacations, concerts, graduations, weddings, and corporate events.
- Nearly everyone creates content for social media.
- Businesses invest heavily in events for experience and digital engagement.
- Platforms such as GuestPix and EventSnap prove willingness to pay for shared event galleries — but primarily solve **collection**.
- EventSphere differentiates by owning the **entire digital experience** of an event in one place: invite → capture → organize → personalize → distribute → reuse.

1.6 Solution summary

Use case	Who	What EventSphere does
Friends	Small groups	Shared temporary gallery, AI albums (people/places/moments), personalized "photos of you", social interactions, no AirDrop/WhatsApp dumps
Celebrations	Hosts + guests	QR access, live uploads, live gallery / photo wall, RSVP, highlight reels, branding, flexible duration
Professional Events	Organizers + attendees + media	Face search at scale, pro photographer albums, media uploads, certificates/badges/resources, auto-delivered personal galleries, centralized digital asset hub

1.7 Product principles

1. **One home** — every digital artifact of an event lives in one place.
2. **Grow with the event** — same product, increasing capability as event complexity grows.
3. **Personal by default** — each person should see *their* memories first.
4. **Build during the event** — galleries assemble live; post-event chase is the anti-pattern.
5. **Privacy-first AI** — face matching and personalization require explicit consent and clear controls.

6. **Complement, don't replace** — camera, photographer, and Instagram remain; EventSphere is the memory layer between them.

1.8 Goals & non-goals

Goals (12 months)

- Guests can join an event in under 30 seconds (QR or link).
- Organizers can create and share an event in under 3 minutes.
- Personalized gallery ("photos of me") available for opted-in users within minutes of upload at Professional scale.
- Reduce "please send photos" coordination for Friends events to near-zero for active groups.
- Support events from 2 people to 10,000+ attendees and 50,000+ media objects.

Non-goals (v1)

- Becoming a general-purpose social network
- Selling physical event merchandise
- Replacing ticketing platforms (e.g., Eventbrite) for paid admission
- Training or selling facial recognition models as a standalone API

1.9 Success metrics

Metric	Definition	Target (MVP → 12 mo)
Time-to-first-upload	Event create → first guest media	< 10 min median
Join conversion	QR open → successful join	> 70%
Personalization coverage	Opted-in attendees with ≥1 face match	> 60% at Pro events
Media findability	Attendees who download/save ≥1 personal photo	> 40% of active guests
Organizer NPS	Post-event survey	> 40
Retention (Friends)	Groups creating ≥2 events / 90 days	> 25%
Gallery completeness	% of guest-captured media that lands in EventSphere vs messaging	Qualitative + survey

1.10 Personas (summary)

Persona	Need	Pain today
Alex — Friend host	One shared album for the weekend trip	WhatsApp chaos, AirDrop failures

Persona	Need	Pain today
Jordan — Wedding couple	Collect candid guest photos + share pro shots	Disposable cameras, Drive folders, endless follow-ups
Sam — Corp event manager	5k photos, certificates, brand assets in one hub	Drive sprawl; attendees can't find themselves
Riley — Attendee	"Just show me photos of me"	Scrolling thousands of images
Casey — Photographer	Deliver client + guest galleries cleanly	Manual sorting and delivery
Morgan — Social / brand	Pull highlight content fast	Assets scattered across teams

1.11 Competitive positioning

Capability	Drive / Dropbox	WhatsApp / AirDrop	GuestPix / EventSnap	Instagram	EventSphere
Guest collection	Partial	Partial	Strong	Weak	Strong
Live gallery / wall	Weak	None	Partial	Weak	Strong
AI people / moments	Weak	None	Weak	Partial	Strong
Personalized "photos of me"	None	None	Weak	Weak	Core
RSVP / invites	None	Partial	Partial	None	Strong (Celebrations+)
Certificates / resources	None	None	None	None	Strong (Professional)
Unified event digital home	No	No	Partial	No	Yes

1.12 Release phasing (recommended)

Phase	Scope
MVP	Friends + Celebrations core: create event, QR join, upload, gallery, likes/comments/downloads, basic AI albums, personalized face gallery (opt-in), 48h Friends expiry, RSVP basic
v1.1	Live photo wall, branding/themes, highlight reel v1, co-organizers
v1.2	Professional: face search at scale, photographer albums, certificates, resource hub, media team roles
v2	Advanced story templates, brand kits, influencer workflows, deeper calendar/CRM integrations

Permission	AG	G	O	CO	P	MT	M	BM	PA
View gallery (approved media)	Y	Y	Y	Y	Y	Y	Y	Y	Y
Upload photos/videos	L	Y	Y	Y	Y	Y	—	L	Y
Like / comment	L	Y	Y	Y	Y	Y	Y	Y	Y
Download media (policy-allowed)	L	Y	Y	Y	Y	Y	—	Y	Y
View personalized gallery	—	Y	Y	Y	Y	Y	—	—	Y
Enable face matching (self)	—	Y	Y	Y	Y	Y	—	—	Y
Create event	—	Y	Y	Y	Y	Y	—	Y	Y
Edit event settings	—	—	F	Y	L	L	—	L	F
Manage RSVP / invitations	—	—	F	Y	—	—	—	—	F
Approve / reject media	—	—	F	Y	L	L	Y	—	F
Delete any media	—	—	F	Y	O	O	Y	—	F
Delete own media	O	O	Y	Y	Y	Y	—	O	Y
Manage roles	—	—	F	L	—	—	—	—	F
Live photo wall control	—	—	F	Y	—	Y	—	—	F
Create photographer album	—	—	Y	Y	Y	Y	—	—	Y
Upload certificates / resources	—	—	F	Y	—	Y	—	Y	F
Download certificates (self)	—	Y	Y	Y	—	—	—	—	Y
Branding / themes	—	—	F	Y	—	—	—	F	F
Export highlight reel / social pack	—	—	F	Y	L	Y	—	F	F
Face search (event-wide admin)	—	—	F	Y	L	L	—	—	F
View analytics	—	—	F	Y	L	L	—	Y	F
Soft-delete / archive event	—	—	F	—	—	—	—	—	F
Hard-delete event + media	—	—	F	—	—	—	—	—	F
Impersonate / support tools	—	—	—	—	—	—	—	—	F

Anonymous guest limits (L)

- Session TTL: 24 hours (configurable; Friends may align to gallery expiry)
- Max uploads per session: 50 media objects
- Must convert to account to enable face matching, cross-device sync, or downloads above soft cap
- Comments may be disabled by organizer for anonymous users

2.4 Role assignment rules

1. Event creator becomes `organizer` automatically.
2. Organizer may invite `co_organizer`, `photographer`, `media_team`, `moderator`, `brand_manager`.
3. Co-organizer may invite `photographer`, `media_team`, `moderator` but **cannot** transfer ownership or hard-delete the event.
4. Ownership transfer requires organizer confirmation + new owner acceptance.
5. Guests join via QR/link/invite; role defaults to `guest`.
6. Elevating a guest to a privileged role requires organizer/co-organizer action.
7. Platform Admin overrides are audited.

2.5 Organization accounts (Professional, future-ready)

Concept	Description
Organization	Company or agency owning many events
Org Admin	Manages members, brand kits, default policies
Org Member	Can create events under the org
External collaborator	Photographer/agency invited per event

v1 may implement single-user organizers with org fields stubbed; schema should support orgs from day one.

2.6 Consent roles (privacy-critical)

Consent	Who grants	Effect
Face embedding consent	Individual user	Enables "photos of me" and face search matching for that user
Guest media license	Uploader (and organizer policy)	Allows event-scoped display/download per event settings
Marketing reuse	Guest opt-in	Allows brand/social export including guest likeness
Parental / minor	Guardian (future)	Restricts face matching and public display

Absence of face consent **must never** block basic gallery viewing or uploading.

3. Feature Catalog

Features are grouped by **platform core** (all use cases) and by **use case** (Friends, Celebrations, Professional). Use cases are capability bundles, not pricing labels.

Priority: **P0** MVP · **P1** v1 · **P2** post-v1

3.1 Platform core (all use cases)

F-CORE-001 Account & identity — P0

- Sign up / sign in: email magic link, Apple, Google
- Phone OTP (optional P1)
- Profile: display name, avatar, locale, timezone
- Session management, logout all devices
- Delete account (GDPR erase cascade)

F-CORE-002 Event create & manage — P0

- Create event: title, cover, date/time range, location, timezone, use-case template
- Event types: friends | celebration | professional
- Edit, archive, restore, hard-delete
- Cover image + short description
- Visibility: private link, invite-only, password, public (flag)

F-CORE-003 Join via QR / link — P0

- Unique event slug + deep link
- QR code generation / download / print sheet
- Passcode gate optional
- Join landing with event preview
- Guest name capture before first upload if anonymous

F-CORE-004 Media upload — P0

- Photo + video upload (camera roll / camera capture)
- Multi-select batch upload
- Background upload with retry
- Client-side compression + EXIF orientation
- Progress UI, pause/resume (P1)
- Duplicate detection (hash)

F-CORE-005 Gallery — P0

- Chronological feed

- Grid / masonry layouts
- Filter: all, photos, videos, people, places, moments, photographer albums
- Infinite scroll / virtualized list
- Media detail viewer (swipe, zoom)
- Sort: newest, oldest, most liked

F-CORE-006 Social interactions — P0

- Like
- Comment (threaded one level P1)
- Download (respect event policy)
- Report content
- Hide from my view (personal)

F-CORE-007 AI organization — P0/P1

- Auto albums by **people, places, moments** (P0 basic clustering; P1 richer)
- Scene/moment labels (cake cutting, speeches, outdoor, etc.) — P1
- Deduplicate near-identical shots — P1
- Quality scoring for highlight candidates — P1

F-CORE-008 Personalized gallery (“Photos of you”) — P0

- Opt-in face enrollment (selfie + consent)
- Private “photos of you” album
- Push when new matches arrive
- Manual “this is / isn’t me” feedback loop
- Revoke consent → delete embeddings + unlink matches

F-CORE-009 Notifications — P0

- In-app + push + email (see Notifications doc)
- Preferences per channel/event

F-CORE-010 Moderation — P0

- Organizer approve-all vs approve-each mode
- Report → review queue
- Hide / remove / ban participant from event
- AI NSFW pre-filter (P1)

F-CORE-011 Search — P0/P1

- Text search: captions, uploader, album names — P0
- Face search (self) — P0
- Face search (admin find attendee) — P1 Professional
- Filter by date/time range, camera type (guest vs pro)

F-CORE-012 Downloads & export — P0

- Single media download
- Bulk zip of selection or personal gallery
- Organizer full export (P1)
- Watermark optional (Professional P1)

F-CORE-013 Analytics (organizer) — P1

- Guests joined, uploads, likes, downloads
 - Top moments, upload velocity during event
 - Personalization opt-in rate
-

3.2 Friends use case

F-FR-001 Temporary private gallery — P0

- Default expiry **48 hours** after event end (or after creation if no end time)
- Countdown banner for remaining time
- Optional extend once (P1) by any member with upload rights — product decision: organizer-only extend

F-FR-002 Frictionless group create — P0

- “Quick event” flow: name + invite friends via share sheet
- No RSVP required
- Default open upload for all members

F-FR-003 Shared dump replacement — P0

- All members upload into one space
- Personalized albums + likes/comments/downloads only what you want
- Auto-reminders: “Add your photos before this gallery expires”

F-FR-004 Lightweight membership — P0

- Share link is enough; accounts encouraged for personalization
 - Max recommended size: ~50 people (soft guidance, not hard block)
-

3.3 Celebrations use case

F-CEL-001 QR-code guest access — P0

- Printable QR, digital QR, table cards

- Multi-language join landing (P1)

F-CEL-002 Live uploads — P0

- Guests upload throughout the event
- Real-time gallery updates (websocket / SSE)

F-CEL-003 Live gallery & photo wall — P0/P1

- Live gallery for mobile/web — P0
- Kiosk / TV photo wall mode — P1 (MVP stretch)
- Display rules: approved only, exclude reported, prefer high-quality

F-CEL-004 Invitations & RSVP — P0

- Invite by link, email, SMS (SMS P1)
- RSVP statuses: going / maybe / declined
- Plus-ones policy
- Guest list export
- Reminder emails

F-CEL-005 Highlight reels & story templates — P1

- AI-generated highlight reel from top moments
- Story-ready vertical templates (export MP4 / images)
- Music bed licensed pack (P2)

F-CEL-006 Branding & themes — P1

- Event colors, fonts (from approved set), cover, logo
- Customizable join page
- Watermark / frame overlays optional

F-CEL-007 Flexible storage & duration — P0

- Organizer sets gallery retention (e.g., 7 / 30 / 90 / 365 days / custom)
- Storage soft quotas with clear upgrade path later (not pitched as tiers in narrative)

3.4 Professional Events use case

F-PRO-001 AI face search at scale — P0/P1

- Attendee self-search — P0
- Organizer/admin lookup by enrolled face or guest identity — P1
- Support 1k–10k+ attendees, 5k–50k+ media

F-PRO-002 Professional photographer albums — P0

- Dedicated albums tagged `source=professional`
- Delivery windows (embargo until organizer publish)
- Separate download policies (full-res for client, web-res for guests)

F-PRO-003 Media team uploads — P1

- Highlight reels, promo clips, press kits
- Role-gated folders
- Approval workflow before public event hub visibility

F-PRO-004 Certificates, badges, resources — P1

- Upload certificate templates or per-attendee files
- Claim / download by authenticated attendee
- Badges (digital) and post-event resource library (PDFs, slides, links)

F-PRO-005 Centralized digital asset hub — P1

- Unified navigation: Photos · Videos · Pro albums · Highlights · Certificates · Resources · Brand kit
- Tagging and collections for media teams

F-PRO-006 Auto-delivered personal galleries — P0

- After event (or rolling during), email/push: “Your photos are ready”
- Deep link into personalized gallery
- One-tap bulk download

F-PRO-007 Domain / SSO access (stretch) — P2

- Corporate SSO (SAML/OIDC)
- Auto-join for verified email domain

3.5 Feature availability by use case

Feature	Friends	Celebrations	Professional
Shared gallery	●	●	●
48h auto-expiry default	●	○	○
Custom retention	○	●	●
QR join	●	●	●
RSVP / invitations	○	●	●
Live photo wall	○	●	●

Feature	Friends	Celebrations	Professional
Themes / branding	○	●	●
Highlight reels	○	●	●
Face personalization	●	●	●
Admin face search	○	○	●
Photographer albums	○	●	●
Certificates / resources	○	○	●
Media team hub	○	○	●
Analytics	Basic	●	●

● = included · ○ = not default (may unlock later without changing use-case story)

3.6 Feature IDs quick index

F-CORE-*, F-FR-*, F-CEL-*, F-PRO-* are stable IDs for tracing into acceptance criteria, tickets, and tests.

4. Screens and Navigation

Screen IDs are stable (S-*). Platforms: **W** Web/PWA · **M** Mobile app · **K** Kiosk/photo wall · **A** Admin.

4.1 Information architecture (primary app)



4.2 Complete screen inventory

Auth & onboarding

ID	Screen	Platforms	Description
S-AUTH-01	Welcome	W/M	Value prop + Get started / Sign in

ID	Screen	Platforms	Description
S-AUTH-02	Sign In	W/M	Email / Apple / Google
S-AUTH-03	Sign Up	W/M	Create account
S-AUTH-04	Magic Link Sent	W/M	Check email state
S-AUTH-05	Magic Link Confirm	W/M	Completes auth
S-ONB-01	Profile Setup	W/M	Name, avatar
S-ONB-02	Face Consent Intro	W/M	Explain personalization; skip allowed
S-ONB-03	Face Enrollment	W/M	Capture selfie(s) for embedding

Home

ID	Screen	Platforms	Description
S-HOME-01	My Events	W/M	Lists of events user belongs to
S-HOME-02	Empty Home	W/M	CTA create or join
S-HOME-03	Notifications Center	W/M	Cross-event notifications
S-HOME-04	Global Search	W/M	Search events & (within accessible) media

Create event

ID	Screen	Platforms	Description
S-CRT-01	Choose Use Case	W/M	Friends / Celebrations / Professional
S-CRT-02	Event Basics	W/M	Title, datetime, location, cover
S-CRT-03	Access & Privacy	W/M	Link/QR, passcode, approval mode
S-CRT-04	Retention & Expiry	W/M	48h Friends default; custom otherwise
S-CRT-05	Invite Guests	W/M	Share link, QR, contacts
S-CRT-06	Created Success	W/M	Share sheet + open hub

Join

ID	Screen	Platforms	Description
S-JOIN-01	Event Landing	W/M	Cover, title, host, CTA Join
S-JOIN-02	Passcode Gate	W/M	Enter event passcode
S-JOIN-03	Guest Name	W/M	For anonymous or first join
S-JOIN-04	Join Success	W/M	Enter gallery / enable face match CTA

Event hub

ID	Screen	Platforms	Description
S-EVT-01	Event Overview	W/M	Summary, countdown, CTAs
S-EVT-02	Gallery Grid	W/M	Primary media browsing
S-EVT-03	Gallery Filters	W/M	Sheet/drawer of filters
S-EVT-04	Album Detail	W/M	People/places/moments/pro album
S-EVT-05	Photos of You	W/M	Personalized gallery
S-EVT-06	Media Viewer	W/M	Fullscreen swipe viewer
S-EVT-07	Comments Sheet	W/M	Likes + comments on media
S-EVT-08	Upload Composer	W/M	Select/capture + upload queue
S-EVT-09	Upload Progress	W/M	Batch progress / failures
S-EVT-10	Guests & RSVP	W/M	List, statuses, invite
S-EVT-11	RSVP Respond	W/M	Going / maybe / declined
S-EVT-12	Moderation Queue	W/M	Approve/reject reported
S-EVT-13	Highlights	W/M	Reels + story exports
S-EVT-14	Highlight Player	W/M	Play generated reel
S-EVT-15	Resources Hub	W/M	PDFs, links, slides (Pro)
S-EVT-16	Certificates	W/M	List + download (Pro)
S-EVT-17	Certificate Detail	W/M	Preview/download
S-EVT-18	Analytics	W/M	Organizer metrics
S-EVT-19	Event Settings	W/M	All event config
S-EVT-20	Roles & Permissions	W/M	Manage team roles
S-EVT-21	Branding / Theme	W/M	Theme editor (Cel/Pro)
S-EVT-22	Share & QR	W/M	Link, QR download, print
S-EVT-23	Face Search (Admin)	W/M	Find attendee photos (Pro)
S-EVT-24	Export Center	W/M	Zips, social packs
S-EVT-25	Danger Zone	W/M	Archive / delete event

Live wall

ID	Screen	Platforms	Description
S-WALL-01	Wall Display	K/W	Full-bleed rotating/collage wall
S-WALL-02	Wall Controller	W/M	Organizer controls (pause, filter)
S-WALL-03	Wall Pairing	W/M	Pair code for kiosk

Profile & legal

ID	Screen	Platforms	Description
S-PROF-01	Profile	W/M	Account overview
S-PROF-02	Edit Profile	W/M	Name, avatar
S-PROF-03	Privacy & Face Data	W/M	Consents, delete embeddings
S-PROF-04	Notification Prefs	W/M	Channel toggles
S-PROF-05	Devices & Sessions	W/M	Active sessions
S-LEGAL-01	Terms	W/M	Terms of service
S-LEGAL-02	Privacy Policy	W/M	Privacy
S-LEGAL-03	Consent Detail	W/M	Face/media consent legalese

System / errors

ID	Screen	Platforms	Description
S-SYS-01	Offline	W/M	Offline banner/full page
S-SYS-02	404 Event	W/M	Event not found / expired
S-SYS-03	403 Forbidden	W/M	No access
S-SYS-04	Maintenance	W/M	Planned downtime
S-SYS-05	Force Update	M	App version too old
S-SYS-06	Report Submitted	W/M	Confirmation

Platform admin (internal)

ID	Screen	Platforms	Description
S-ADM-01	Admin Home	A	Ops dashboard
S-ADM-02	User Lookup	A	Support tools
S-ADM-03	Event Lookup	A	Support tools
S-ADM-04	Abuse Queue	A	Platform-level reports

4.3 Navigation map

Global (authenticated)

From	Action	To
Any	Tap logo / Home tab	S-HOME-01

From	Action	To
S-HOME-01	Tap event card	S-EVT-01
S-HOME-01	Tap + Create	S-CRT-01
S-HOME-01	Tap Join / scan	Camera → S-JOIN-01
S-HOME-01	Profile avatar	S-PROF-01
S-HOME-01	Bell	S-HOME-03

Event hub tabs (mobile)

Tab	Screen
Home	S-EVT-01
Gallery	S-EVT-02
Upload	S-EVT-08
Guests	S-EVT-10 (hidden for Friends if RSVP off)
More	Settings, Highlights, Resources, Analytics...

Media viewer

From	Action	To
S-EVT-02 / 04 / 05	Tap media	S-EVT-06
S-EVT-06	Swipe	Next/prev media
S-EVT-06	Comment icon	S-EVT-07
S-EVT-06	Close	Previous gallery

Join deep links

URL pattern	Screen
/e/{slug}	S-JOIN-01
/e/{slug}/join	S-JOIN-01 → join API
/e/{slug}/wall	S-WALL-01
/e/{slug}/rsvp	S-EVT-11
/m/{mediaId}	S-EVT-06 (if authorized)

4.4 Role-based screen visibility

Screen	Guest	Organizer	Photographer	Media	Anon
Gallery	●	●	●	●	●
Photos of You	●*	●*	●*	●*	—
Upload	●	●	●	●	●
Moderation	—	●	○	○	—
Guests/RSVP	view self	●	—	—	—
Analytics	—	●	○	○	—
Settings	—	●	○	○	—
Certificates	●	●	—	●	—
Admin face search	—	●	○	○	—
Wall controller	—	●	—	●	—

* Requires face consent + enrollment

○ = if granted by organizer

4.5 Navigation rules

1. Deep links must land on join/landing if user is not a member; after join, continue to intended screen.
2. Expired Friends galleries show S-SYS-02 with "Gallery expired" and option to request organizer extend (if enabled).
3. Back stack on mobile: Media Viewer is a stack modal; closing returns to exact scroll position.
4. Kiosk wall mode hides all chrome; exit requires long-press + passcode (organizer-set).
5. Unauthenticated users can complete Join → Gallery → Upload without account until face match or bulk download requires auth.

5. Complete User Flows

Flows use ID `UF-*`. Each step lists actor, screens, and system actions.

UF-01 Create Friends event and share

Actor: Organizer

Goal: Spin up a temporary shared gallery in < 3 minutes

1. Open app → S-HOME-01 → Create
2. Choose **Friends** (S-CRT-01)
3. Enter title, optional place/time (S-CRT-02)
4. Confirm privacy: private link; uploads open; retention 48h (S-CRT-03/04) — prefilled
5. System creates event + QR + slug
6. Share sheet (S-CRT-05/06): Messages, WhatsApp, Copy link, Show QR
7. Land on Event Overview (S-EVT-01)

Success: Event exists; at least one share action available; countdown visible.

UF-02 Guest joins via QR and uploads

Actor: Guest (may be anonymous)

Goal: Join and contribute photos during the event

1. Scan QR → S-JOIN-01
2. Optional passcode → S-JOIN-02
3. Tap Join → if no account: S-JOIN-03 enter display name → session created
4. S-JOIN-04 → CTA Upload / View gallery
5. S-EVT-08 select media → upload queue (S-EVT-09)
6. System: virus/MIME check → store → process variants → AI enqueue
7. Media appears in gallery (pending if approval required)

Success: Media visible to authorized viewers per moderation policy.

UF-03 Personalized “photos of you”

Actor: Authenticated guest

Goal: See only photos containing them

1. From S-JOIN-04 or S-EVT-01 CTA “Find photos of me”
2. S-ONB-02 consent explanation → Accept
3. S-ONB-03 capture 1–3 selfies

4. System creates face embedding; stores consent record
5. Matcher runs against event media (async)
6. S-EVT-05 populates; push "We found N photos of you"
7. Guest can mark false positives → model feedback

Revocation path: S-PROF-03 → Revoke → embeddings deleted; album cleared; re-processing stops.

UF-04 Celebrations: invite, RSVP, live wall

Actor: Organizer + Guests

Goal: Run a wedding/birthday with live collection

Organizer

1. Create **Celebrations** event (S-CRT-*)
2. Configure RSVP + branding (S-EVT-19/21)
3. Send invites (S-EVT-10) / print QR (S-EVT-22)
4. Launch photo wall on venue display (S-WALL-01) via pair code (S-WALL-03)

Guest

1. Open invite link → S-EVT-11 RSVP
2. At event, scan QR → join → upload (UF-02)
3. Approved media streams to wall within target latency

Success: RSVP recorded; live wall shows approved media; guests can browse gallery on phone.

UF-05 Professional: 5,000 photos → personal delivery

Actor: Organizer, Photographer, Attendees

Goal: Replace Drive-folder hell

1. Organizer creates **Professional** event; adds photographer role
2. Photographer uploads pro album (possibly embargoed)
3. Guests join (domain or QR) and optionally enroll face
4. During/after event, media processed; face matches computed
5. Organizer publishes pro album
6. System sends "Your photos are ready" to opted-in attendees (email/push)
7. Attendee opens personalized gallery → bulk download
8. Certificates/resources available in hub

Success: Attendee finds their photos without scrolling the full set; download completes.

UF-06 Moderation of inappropriate content

1. Guest reports media from S-EVT-06
 2. Media flagged; optionally auto-hidden if AI NSFW score high (P1)
 3. Organizer/moderator reviews S-EVT-12
 4. Actions: Approve, Remove, Ban uploader
 5. Reporter receives S-SYS-06 / notification of resolution (no details of other users)
-

UF-07 Highlight reel generation

1. Organizer opens S-EVT-13 → Generate highlight
 2. System selects candidates via quality + moments + diversity
 3. Job status: queued → processing → ready / failed
 4. Organizer previews S-EVT-14 → edit exclusions (P1) → export
 5. Share to social / download MP4
-

UF-08 Event expiry (Friends 48h)

1. At T-6h: notify members "Gallery expires in 6 hours — download what you love"
 2. At T-0: event state → `expired`; uploads blocked; gallery read-only for grace period (24h default)
OR immediate opaque expiry per config
 3. After retention: media hard-deleted; event tombstoned
 4. Organizer may extend before expiry if policy allows (once)
-

UF-09 Co-organizer collaboration

1. Organizer invites email as co-organizer (S-EVT-20)
 2. Invitee accepts → role granted
 3. Co-organizer can moderate, edit settings (non-destructive), manage guests
 4. Cannot hard-delete or transfer ownership
-

UF-10 Account deletion (GDPR)

1. User requests delete in S-PROF-01
 2. Confirm password/magic reauth
 3. System schedule: revoke sessions → delete embeddings → anonymize comments → remove owned media or transfer per policy → delete profile
 4. Organizer-owned events: require transfer or delete events first (blocking error)
-

UF-11 Offline upload resilience

1. Guest selects media offline
 2. Queue persisted locally
 3. On reconnect, uploads resume with idempotency keys
 4. Conflicts: if event expired while offline, show error and keep local copy note
-

UF-12 Certificate claim

1. Attendee opens S-EVT-16
 2. System matches attendee identity to certificate record
 3. Preview S-EVT-17 → download PDF
 4. If not eligible: error `CERT_NOT_ISSUED`
-

5.1 Flow → screen trace matrix (primary)

Flow	Key screens
UF-01	S-CRT-01...06, S-EVT-01
UF-02	S-JOIN-01...04, S-EVT-08/09/02
UF-03	S-ONB-02/03, S-EVT-05
UF-04	S-EVT-10/11/22, S-WALL-*
UF-05	S-EVT-05/16/24, notifications
UF-06	S-EVT-06/12
UF-07	S-EVT-13/14
UF-08	S-EVT-01, notifications, S-SYS-02
UF-09	S-EVT-20
UF-10	S-PROF-01
UF-11	S-EVT-08/09, S-SYS-01
UF-12	S-EVT-16/17

6. State Diagrams

6.1 Event lifecycle

```
stateDiagram-v2
    [*] --> draft: create
    draft --> scheduled: publish / share
    draft --> cancelled: cancel
    scheduled --> live: start_at reached OR organizer force-live
    live --> ended: end_at reached OR organizer end
    ended --> expired: retention elapsed (Friends default 48h)
    ended --> archived: organizer archive
    expired --> archived: tombstone
    archived --> [*]: hard delete
    cancelled --> [*]: hard delete after grace
    ended --> live: organizer reopen (policy)
    expired --> live: organizer extend (before purge)
```

State	Uploads	Gallery view	Notes
draft	Organizer only	Organizer only	Not joinable via public QR until published
scheduled	Per settings	Members	Pre-event
live	Yes	Members	Real-time wall active
ended	Optional grace	Members	Default: uploads still allowed until expiry
expired	No	Grace read-only or none	Friends default path
archived	No	Organizer	Cold storage optional
cancelled	No	No	Invites invalidated

6.2 Media object lifecycle

```
stateDiagram-v2
    [*] --> uploading: client starts
    uploading --> stored: bytes accepted
    uploading --> failed_upload: error
    stored --> processing: pipeline
    processing --> ready: variants + metadata done
    processing --> failed_processing: error
    ready --> pending_approval: if moderation required
    ready --> published: auto-approve mode
    pending_approval --> published: approve
    pending_approval --> rejected: reject
    published --> flagged: report / AI flag
    flagged --> published: dismiss
    flagged --> removed: remove
    published --> removed: organizer/user delete
    rejected --> [*]: purge
    removed --> [*]: purge after retention
    failed_upload --> [*]
    failed_processing --> stored: retry
```

6.3 Face enrollment & matching

```
stateDiagram-v2
    [*] --> not_enrolled
    not_enrolled --> consent_shown: open CTA
    consent_shown --> not_enrolled: decline / skip
    consent_shown --> capturing: accept
    capturing --> enrolled: embedding saved
    capturing --> not_enrolled: cancel
    enrolled --> matching: new media / backfill
    matching --> enrolled: matches updated
    enrolled --> revoked: user revokes
    revoked --> not_enrolled: embeddings deleted
```

6.4 RSVP

```
stateDiagram-v2
    [*] --> invited
    invited --> going
    invited --> maybe
    invited --> declined
    going --> maybe
    going --> declined
    maybe --> going
    maybe --> declined
    declined --> going
    declined --> maybe
    going --> checked_in: optional onsite
    checked_in --> [*]
```

6.5 Highlight / export job

```
stateDiagram-v2
    [*] --> queued
    queued --> running
    running --> succeeded
    running --> failed
    failed --> queued: retry
    succeeded --> [*]
```

6.6 Upload batch (client)

```
stateDiagram-v2
    [*] --> queued_local
    queued_local --> uploading
    uploading --> succeeded
    uploading --> failed
    failed --> queued_local: retry
    uploading --> paused: user/network
    paused --> uploading: resume
    succeeded --> [*]
```

6.7 Membership

```
stateDiagram-v2
    [*] --> none
    none --> pending_invite: invited
    none --> active: join via QR/link
    pending_invite --> active: accept
    pending_invite --> none: decline / expire
    active --> banned: moderator ban
    active --> left: user leave
    banned --> [*]
    left --> active: rejoin if allowed
```

6.8 Notification delivery

```
stateDiagram-v2
    [*] --> created
    created --> suppressed: user prefs / quiet hours
    created --> sending
    sending --> delivered
    sending --> failed
    failed --> sending: retry ≤ N
    failed --> dead: letter
    delivered --> read: user opens
    suppressed --> [*]
    dead --> [*]
    read --> [*]
```

7. Business Rules

IDs: BR-*. Rules are normative for engineering and QA.

7.1 Event rules

ID	Rule
BR-EVT-001	Every event has exactly one <code>organizer</code> (owner) at all times.
BR-EVT-002	Use case is set at creation: <code>friends</code> <code>celebration</code> <code>professional</code> ; changing use case after publish is allowed only before first guest join, else blocked.
BR-EVT-003	Friends events default <code>retention_hours</code> = 48 after <code>end_at</code> (or <code>created_at</code> if no end).
BR-EVT-004	Celebrations/Professional default retention is organizer-selected; minimum 7 days; maximum 3650 days (10 years) unless legal hold.
BR-EVT-005	Expired events reject new uploads with <code>EVENT_EXPIRED</code> .
BR-EVT-006	Hard delete is irreversible after confirmation; soft delete/archive precedes purge job.
BR-EVT-007	Public discoverability default = false.
BR-EVT-008	Passcode, if set, is required for all non-privileged joins.
BR-EVT-009	Event slug is immutable after first publish.
BR-EVT-010	Live wall may only display <code>published</code> media.

7.2 Membership & roles

ID	Rule
BR-MEM-001	Joining creates an <code>event_membership</code> with role <code>guest</code> unless invite specifies otherwise.
BR-MEM-002	Anonymous sessions cannot enroll face data.
BR-MEM-003	Co-organizer cannot remove the organizer or assign another organizer.

ID	Rule
BR-MEM-004	Banned users cannot rejoin unless unbanned by organizer/co-organizer.
BR-MEM-005	Leaving an event does not delete the user's prior uploads (they remain unless user deletes own media).
BR-MEM-006	Role elevation is audited.

7.3 Media rules

ID	Rule
BR-MED-001	Allowed image types: JPEG, PNG, WEBP, HEIC. Video: MP4, MOV.
BR-MED-002	Max image size: 50 MB. Max video size: 500 MB (Friends may lower to 100 MB). Max video duration: 5 minutes (P0).
BR-MED-003	Uploader always retains right to delete own media.
BR-MED-004	In <code>approve_each</code> mode, media is invisible to general guests until approved.
BR-MED-005	Organizer/co-organizer/moderator can remove any media.
BR-MED-006	Duplicate content-hash within same event is rejected or linked as duplicate (config: reject).
BR-MED-007	EXIF GPS may be stripped by default for Friends; Celebrations/Pro: organizer policy (<code>keep</code> <code>strip</code>).
BR-MED-008	Download policy: <code>all</code> <code>members</code> <code>organizer_only</code> <code>disabled</code> .
BR-MED-009	Comments disabled globally if organizer sets <code>comments_enabled=false</code> .
BR-MED-010	Removed media is omitted from AI albums, walls, and personal galleries within 60 seconds of removal.

7.4 AI & face data

ID	Rule
BR-AI-001	Face embeddings require explicit consent (<code>face_matching = true</code>).

ID	Rule
BR-AI-002	Embeddings are per-user, encrypted at rest, never exposed via API.
BR-AI-003	Revoking consent deletes embeddings within 24 hours and unlinks matches.
BR-AI-004	Face matching is event-scoped; embeddings are not used to identify users across unrelated events without separate consent flag (<code>cross_event_matching</code> , default false).
BR-AI-005	Minors: if <code>age_gate</code> indicates under 16 (or local digital consent age), face matching is disabled.
BR-AI-006	Admin face search (Professional) may search media, not export raw biometric templates.
BR-AI-007	AI album generation must degrade gracefully: if AI fails, chronological gallery remains fully usable.
BR-AI-008	Highlight reel must include only <code>published</code> media and respect marketing-consent flags when exported for public social use.

7.5 RSVP & invitations

ID	Rule
BR-RSVP-001	RSVP exists only when <code>rsvp_enabled=true</code> (Celebrations/Pro default on; Friends off).
BR-RSVP-002	Plus-ones cannot exceed <code>max_plus_ones</code> per invite.
BR-RSVP-003	Declined guests may still join gallery if <code>allow_declined_join=true</code> (default true for Celebrations).
BR-RSVP-004	Invite tokens expire after <code>invite_ttl_days</code> (default 90) or event archive.

7.6 Notifications

ID	Rule
BR-NOT-001	Transactional notifications (security, expiry) ignore marketing mute but respect hard bounce.
BR-NOT-002	“New photos of you” requires face consent + notification preference.
BR-NOT-003	Quiet hours default 22:00–08:00 local; expiry warnings still send.

7.7 Retention & deletion

ID	Rule
BR-RET-001	Friends: purge media at retention end; metadata tombstone retained 30 days for support.
BR-RET-002	User account deletion removes embeddings immediately; media owned solely as guest uploads deleted or anonymized per regional policy.
BR-RET-003	Legal hold on Professional events blocks purge.
BR-RET-004	Exports (zips) are ephemeral; delete within 72 hours of generation.

7.8 Rate limits (product-level)

ID	Rule
BR-RATE-001	Anonymous: max 50 uploads / session / event.
BR-RATE-002	Authenticated guest: max 500 uploads / event / day (soft); burst limited.
BR-RATE-003	Comments: max 30 / user / event / hour.
BR-RATE-004	Report: max 20 / user / day.
BR-RATE-005	Create event: max 20 / user / day.

7.9 Branding & export

ID	Rule
BR-BRAND-001	Custom fonts limited to approved font catalog (license compliance).
BR-BRAND-002	Social export packs exclude guests who opted out of marketing reuse when pack type = <code>public_social</code> .
BR-BRAND-003	Internal recap packs may include all published media for organizer download.

8. Validation Rules, Error Messages, Edge Cases & Exception Handling

8.1 Field validation

User

Field	Rules	Error code	Message
email	Required, RFC 5322, max 320	VAL_EMAIL_INVALID	Enter a valid email address.
display_name	Required on join, 1–40 chars, trimmed, no control chars	VAL_NAME_INVALID	Name must be 1–40 characters.
password	If password auth used: 8–128, not breached (P1)	VAL_PASSWORD_WEAK	Choose a stronger password (at least 8 characters).
locale	BCP 47	VAL_LOCALE_INVALID	Unsupported language setting.

Event

Field	Rules	Error code	Message
title	Required, 1–80 chars	VAL_TITLE_REQUIRED	Give your event a name.
use_case	enum	VAL_USE_CASE	Choose Friends, Celebrations, or Professional Events.
start_at / end_at	end ≥ start; start not > 5 years future	VAL_DATE_RANGE	End time must be after start time.
slug	lowercase kebab, 3–48, unique	VAL_SLUG	That link is unavailable. Try another.
passcode	Optional 4–12 alphanumeric	VAL_PASSCODE	Passcode must be 4–12 letters or numbers.
retention_hours	Friends default 48; others per BR-EVT-004	VAL_RETENTION	Choose a valid gallery duration.
max_guests	Optional positive int ≤ 100000	VAL_MAX_GUESTS	Guest limit is too high.

Media

Field	Rules	Error code	Message
content_type	Allow-list only	VAL_MEDIA_TYPE	This file type isn't supported.
size	Per BR-MED-002	VAL_MEDIA_SIZE	This file is too large.
duration	Video ≤ max	VAL_MEDIA_DURATION	Videos can be up to 5 minutes.

Field	Rules	Error code	Message
caption	Max 500 chars	VAL_CAPTION	Caption is too long.

RSVP

Field	Rules	Error code	Message
status	going maybe declined	VAL_RSVP_STATUS	Choose a valid RSVP response.
plus_ones	0..max	VAL_PLUS_ONES	You've exceeded the plus-one limit.

Face enrollment

Field	Rules	Error code	Message
consent	Must be true	VAL_FACE_CONSENT	To find photos of you, please accept face matching.
selfie quality	Face detected, min resolution, not blurry	VAL_FACE_QUALITY	We couldn't detect a clear face. Try better lighting.
selfie count	1-3	VAL_FACE_COUNT	Upload between 1 and 3 selfies.

8.2 API / domain error catalog

Code	HTTP	When	User-facing message
AUTH_REQUIRED	401	Missing/expired session	Sign in to continue.
AUTH_FORBIDDEN	403	Lacks permission	You don't have access to do that.
EVENT_NOT_FOUND	404	Bad slug/id	We can't find this event.
EVENT_EXPIRED	410	Past retention	This gallery has expired.
EVENT_CANCELLED	410	Cancelled	This event was cancelled.
EVENT_FULL	409	max_guests reached	This event is full.
PASSCODE_INVALID	403	Wrong passcode	That passcode isn't correct.
MEMBERSHIP_BANNED	403	Banned	You can no longer join this event.
MEDIA_NOT_FOUND	404	Deleted/missing	This photo is no longer available.
MEDIA_PENDING	403	Not approved yet	This photo is waiting for approval.
UPLOAD_QUOTA	429	Rate/quota	You've reached the upload limit. Try again later.
DUPLICATE_MEDIA	409	Same hash	This photo was already uploaded.
FACE_NOT_ENROLLED	400	Action needs enrollment	Set up photo matching first.

Code	HTTP	When	User-facing message
FACE_DISABLED_MINOR	403	Age gate	Photo matching isn't available for this account.
CERT_NOT_ISSUED	404	No cert	No certificate is available for you yet.
JOB_FAILED	500	Highlight/export fail	Something went wrong creating your highlight. Try again.
STORAGE_UNAVAILABLE	503	Upstream	Uploads are temporarily unavailable.
VALIDATION_FAILED	422	Field errors	Check the highlighted fields.
RATE_LIMITED	429	Generic	You're going too fast. Please wait a moment.
CONFLICT_STATE	409	Invalid transition	This action isn't available in the event's current state.
OWNERSHIP_BLOCK	409	Delete account while owning events	Transfer or delete your events before deleting your account.

8.3 Edge cases

ID	Scenario	Expected behavior
EC-001	Two users upload same photo	Second gets <code>DUPLICATE_MEDIA</code> or silent link; gallery shows one
EC-002	Organizer deletes media while guest views it	Viewer shows "no longer available"; wall skips
EC-003	Guest enrolls face after 3k photos exist	Backfill job; progressive results in Photos of You
EC-004	QR scanned after expiry	S-SYS-02 expired state; no join
EC-005	Clock skew on client upload start_at	Server time authoritative
EC-006	HEIC on web browser without support	Transcode server-side; show processing state
EC-007	Live wall network drop	Last frame + reconnect; no crash blank forever (>3s spinner then cached collage)
EC-008	Organizer demotes self incorrectly	Blocked; must transfer ownership first
EC-009	RSVP going but never joins	Still counted in RSVP analytics; not in "active guests"
EC-010	Face false positive	User taps "Not me" → unlink + suppress similar
EC-011	Partial batch upload failure	Per-file errors; succeeded files remain

ID	Scenario	Expected behavior
EC-012	Event use case change after guests joined	Rejected with <code>CONFLICT_STATE</code>
EC-013	Download zip for 10k personal photos	Async job + email link; not synchronous
EC-014	Anonymous session on second device	New session; cannot see prior anonymous uploads as "mine" unless claimed
EC-015	Claim anonymous uploads after signup	If same device session token present, attach uploads to account
EC-016	Photographer embargo	Guests cannot see pro album until publish
EC-017	Marketing export with mixed consents	Filter non-consenting likenesses for public packs
EC-018	Simultaneous approve + uploader delete	Delete wins; moderation item closes
EC-019	Storage provider outage mid-upload	Client retries; idempotent complete
EC-020	User in 2 photos of 5000	Personalized gallery shows 2; full gallery still available

8.4 Exception handling requirements

1. **Client:** All mutations idempotent via `Idempotency-Key` header for uploads and payments (future).
2. **API:** Never leak stack traces; return `request_id` on 5xx.
3. **Jobs:** At-least-once processing with idempotent handlers; poison queue after 5 failures.
4. **AI pipeline:** Failure must not block `ready` state for basic gallery variants; AI fields nullable.
5. **Partial outages:** Read-only mode flag for gallery if uploads disabled.
6. **Support:** Platform admin can retrieve `request_id` → logs without accessing face embeddings raw values.

8.5 Empty & loading UX copy

State	Copy
Empty gallery	"No photos yet. Be the first to add a memory."
Empty Photos of You (enrolled)	"We're still looking — check back as more photos are added."
Empty Photos of You (not enrolled)	"Turn on photo matching to find pictures of you automatically."
Empty guests	"Share the QR code so guests can join."
Expired	"This gallery has closed. Hope you saved your favorites."

State	Copy
Moderation empty	"You're all caught up."
Search no results	"No matches. Try another filter."

9. API Contracts

9.1 Conventions

Item	Standard
Style	REST JSON over HTTPS
Base URL	<code>https://api.eventsphere.app/v1</code>
Auth	<code>Authorization: Bearer <access_token></code>
Idempotency	<code>Idempotency-Key</code> on POST uploads & jobs
Time	ISO-8601 UTC
IDs	ULID/UUID strings
Pagination	Cursor: <code>?cursor=&limit=</code> (default 30, max 100)
Errors	<code>{ "error": { "code", "message", "details?", "request_id" } }</code>
Realtime	WebSocket <code>wss://api.eventsphere.app/v1/ws</code> or SSE <code>/events/{id}/stream</code>

Anonymous guest sessions use short-lived bearer tokens issued by join endpoints.

9.2 Auth

POST `/auth/magic-link`

Request: `{ "email": "a@b.com" }`

Response 202: `{ "ok": true }`

POST `/auth/magic-link/confirm`

Request: `{ "token": "..."`

Response 200: `{ "access_token", "refresh_token", "user" }`

POST `/auth/oauth/{provider}`

Providers: `apple, google`

Request: `{ "id_token": "..."`

Response: `tokens + user`

POST `/auth/refresh`

Request: `{ "refresh_token" }` → new tokens

POST /auth/logout

Invalidates refresh token / session

9.3 Users

GET /me

Response: User object

PATCH /me

Body: { "display_name?", "avatar_media_id?", "locale?", "timezone?" }

DELETE /me

Starts account deletion; may return 409 OWNERSHIP_BLOCK

GET /me/consents

PUT /me/consents

Body: { "face_matching": true, "cross_event_matching": false, "marketing_reuse_default": false }

9.4 Events

POST /events

```
{
  "title": "Amritsar Trip",
  "use_case": "friends",
  "description": "Weekend",
  "start_at": "2026-08-01T10:00:00Z",
  "end_at": "2026-08-03T18:00:00Z",
  "timezone": "Asia/Karachi",
  "location": { "name": "Amritsar", "lat": null, "lng": null },
  "cover_media_id": null,
  "access": {
    "visibility": "private_link",
    "passcode": null,
    "rsvp_enabled": false,
    "approval_mode": "auto",
    "download_policy": "members",
    "comments_enabled": true
  },
  "retention_hours": 48
}
```

Response 201: Event object including slug, qr_url, join_url

GET /events

List my events: ?filter=upcoming|live|past

GET /events/{eventId}

GET /events/by-slug/{slug}

Public-safe landing payload if not a member

PATCH /events/{eventId}

Organizer/co-organizer

POST /events/{eventId}/publish

draft → scheduled/live

POST /events/{eventId}/end

POST /events/{eventId}/extend

Body: { "additional_hours": 48 } (Friends policy)

POST `/events/{eventId}/archive`

DELETE `/events/{eventId}`

Hard delete (owner)

9.5 Join & membership

POST `/events/by-slug/{slug}/join`

```
{
  "passcode": "optional",
  "display_name": "Riley",
  "as_anonymous": true
}
```

Response: { "membership", "access_token?", "event" }

GET `/events/{eventId}/members`

PATCH `/events/{eventId}/members/{userId}`

Body: { "role": "photographer" }

DELETE `/events/{eventId}/members/{userId}`

Remove or leave (/me variant)

POST `/events/{eventId}/members/{userId}/ban`

POST `/events/{eventId}/members/{userId}/unban`

9.6 Invitations & RSVP

POST `/events/{eventId}/invitations`

```
{ "emails": ["a@b.com"], "role": "guest", "message": "Join us!" }
```

POST /invitations/{token}/accept

POST /events/{eventId}/rsvp

Body: { "status": "going", "plus_ones": 1 }

GET /events/{eventId}/rsmps

9.7 Media

POST /events/{eventId}/media/uploads

Init multipart / direct-to-storage:

```
{
  "filename": "IMG_01.HEIC",
  "content_type": "image/heic",
  "byte_size": 2345678,
  "checksum_sha256": "...",
  "captured_at": "2026-08-01T12:00:00Z",
  "client_idempotency_key": "..."
}
```

Response: { "media_id", "upload_url", "upload_headers", "expires_at" }

POST /events/{eventId}/media/{mediaId}/complete

Confirms upload; enqueues processing

GET /events/{eventId}/media

Query: album, person_id, moment, source=guest|professional, cursor

GET /media/{mediaId}

DELETE /media/{mediaId}

POST /media/{mediaId}/like

DELETE /media/{mediaId}/like

GET /media/{mediaId}/comments

POST /media/{mediaId}/comments

Body: { "body": "Love this!" }

POST /media/{mediaId}/report

Body: { "reason": "spam|nsfw|harassment|other", "notes?" }

POST /media/{mediaId}/approve

POST /media/{mediaId}/reject

9.8 Albums & personalization

GET /events/{eventId}/albums

GET /events/{eventId}/albums/{albumId}/media

POST /me/face-enrollment

Multipart selfies + { "consent": true } Response: { "enrollment_id", "status": "enrolled" }

DELETE /me/face-enrollment

GET /events/{eventId}/me/photos

Personalized gallery

POST /events/{eventId}/me/photos/{mediaId}/feedback

Body: { "is_me": false }

POST /events/{eventId}/face-search (Professional admin)

Body: { "target_user_id": "..." } OR { "enrollment_selfie_media_id": "..." }

Response: media list

9.9 Highlights, exports, wall

POST /events/{eventId}/highlights

Body: { "style": "recap_30s", "music_track_id?": null }

Response: { "job_id" }

GET /jobs/{jobId}

GET /events/{eventId}/highlights

POST /events/{eventId}/exports

Body: { "type": "personal_zip|event_zip|social_pack", "media_ids?": [] }

Response: { "job_id" }

GET /events/{eventId}/wall/config

PATCH /events/{eventId}/wall/config

GET /events/{eventId}/wall/stream (SSE)

9.10 Resources & certificates (Professional)

POST /events/{eventId}/resources

GET /events/{eventId}/resources

DELETE /resources/{resourceId}

POST /events/{eventId}/certificates

Bulk issue metadata:

```
{
  "items": [
    { "user_id": "...", "title": "Attendance", "file_media_id": "..." }
  ]
}
```

GET /events/{eventId}/me/certificates

GET /certificates/{certificateId}/download → signed URL

9.11 Analytics

GET /events/{eventId}/analytics/summary

```
{
  "guests_joined": 420,
  "media_count": 5120,
  "likes": 9800,
  "downloads": 1500,
  "face_opt_in_rate": 0.62,
  "upload_velocity": [{ "t": "...", "count": 120 }]
}
```

9.12 Core resource shapes

User

```
{
  "id": "user_...",
  "email": "a@b.com",
  "display_name": "Riley",
  "avatar_url": "https://...",
  "locale": "en",
  "timezone": "Asia/Karachi",
  "created_at": "..."
}
```

Event

```
{
  "id": "evt_...",
  "slug": "amritsar-trip",
  "title": "Amritsar Trip",
  "use_case": "friends",
  "state": "live",
  "start_at": "...",
  "end_at": "...",
  "retention_hours": 48,
  "expires_at": "...",
  "join_url": "https://eventsphere.app/e/amritsar-trip",
  "qr_url": "https://...",
  "cover_url": "https://...",
  "access": { "visibility": "private_link", "approval_mode": "auto", "rsvp_enabled": false },
  "counts": { "members": 5, "media": 120 },
  "viewer_membership": { "role": "organizer", "status": "active" }
}
```


Media

```
{
  "id": "med_...",
  "event_id": "evt_...",
  "type": "image",
  "state": "published",
  "source": "guest",
  "urls": {
    "thumb": "...",
    "md": "...",
    "lg": "...",
    "original": "..."
  },
  "width": 4032,
  "height": 3024,
  "duration_ms": null,
  "caption": "",
  "uploader": { "id": "user_...", "display_name": "Alex" },
  "like_count": 3,
  "liked_by_me": false,
  "comment_count": 1,
  "captured_at": "...",
  "created_at": "..."
}
```

9.13 Webhooks (organizer integrations — P2)

`media.published`, `rsvp.updated`, `event.expired`, `export.ready`
Signed with `X-EventSphere-Signature: sha256=...`

9.14 Status code summary

Status	Meaning
200/201/202	OK / created / accepted async
400	Bad request
401	Auth required
403	Forbidden
404	Not found
409	Conflict
410	Gone (expired/cancelled)
422	Validation
429	Rate limit
500	Server error

Status	Meaning
503	Dependency unavailable

10. Database Schema

Logical relational model (PostgreSQL-oriented). IDs are `uuid` unless noted. Timestamps `timestampz`. Soft deletes via `deleted_at` where specified.

10.1 ER overview

```
erDiagram
    users ||--o{ event_memberships : has
    users ||--o{ events : owns
    events ||--o{ event_memberships : has
    events ||--o{ media_objects : contains
    events ||--o{ albums : has
    albums ||--o{ album_media : contains
    media_objects ||--o{ album_media : in
    media_objects ||--o{ media_likes : has
    media_objects ||--o{ media_comments : has
    users ||--o{ face_enrollments : has
    face_enrollments ||--o{ face_matches : produces
    media_objects ||--o{ face_matches : matched
    events ||--o{ invitations : sends
    events ||--o{ rsvps : tracks
    events ||--o{ resources : has
    events ||--o{ certificates : has
    events ||--o{ jobs : runs
    users ||--o{ devices : has
    users ||--o{ notification_prefs : has
```

10.2 Tables

users

Column	Type	Notes
id	uuid PK	
email	citext UNIQUE	nullable for pure anon upgrade path
display_name	text NOT NULL	
avatar_url	text	
locale	text DEFAULT 'en'	
timezone	text	
role_global	text	user platform_admin
age_gate	text	unknown adult minor
created_at	timestampz	

Column	Type	Notes
deleted_at	timestamptz	

auth_identities

Column	Type	Notes
id	uuid PK	
user_id	uuid FK users	
provider	text	email apple google
provider_subject	text	
UNIQUE(provider, provider_subject)		

sessions

Column	Type	Notes
id	uuid PK	
user_id	uuid FK nullable	null for anonymous
anonymous_key	text	
event_id	uuid FK nullable	anon scoped
refresh_hash	text	
expires_at	timestamptz	
created_at	timestamptz	
revoked_at	timestamptz	

organizations (future-ready)

Column	Type	Notes
id	uuid PK	
name	text	
created_at	timestamptz	

organization_members

Column	Type	Notes
org_id	uuid FK	
user_id	uuid FK	
org_role	text	admin member
PK(org_id, user_id)		

events

Column	Type	Notes
id	uuid PK	
org_id	uuid FK nullable	
owner_user_id	uuid FK users	
slug	citext UNIQUE	
title	text	
description	text	
use_case	text	friends celebration professional
state	text	see state diagram
start_at	timestampz	
end_at	timestampz	
timezone	text	
location_name	text	
location_lat	numeric	
location_lng	numeric	
cover_media_id	uuid	
visibility	text	
passcode_hash	text	
rsvp_enabled	bool	
approval_mode	text	auto approve_each
download_policy	text	
comments_enabled	bool	
strip_exif_gps	bool	
retention_hours	int	
expires_at	timestampz	
max_guests	int	
allow_declined_join	bool	
branding	jsonb	theme config
wall_config	jsonb	
legal_hold	bool DEFAULT false	
created_at	timestampz	
updated_at	timestampz	

Column	Type	Notes
deleted_at	timestampz	

Indexes: (state, expires_at), (owner_user_id), (org_id)

event_memberships

Column	Type	Notes
id	uuid PK	
event_id	uuid FK	
user_id	uuid FK nullable	
session_id	uuid FK nullable	anon
role	text	
status	text	active banned left
display_name_override	text	
joined_at	timestampz	
UNIQUE(event_id, user_id) WHERE user_id IS NOT NULL		

invitations

Column	Type	Notes
id	uuid PK	
event_id	uuid FK	
email	citext	
role	text	
token_hash	text UNIQUE	
expires_at	timestampz	
accepted_at	timestampz	
created_by	uuid FK	

rsvps

Column	Type	Notes
id	uuid PK	
event_id	uuid FK	
user_id	uuid FK nullable	
email	citext	

Column	Type	Notes
status	text	
plus_ones	int	
checked_in_at	timestampz	
UNIQUE(event_id, email)		

media_objects

Column	Type	Notes
id	uuid PK	
event_id	uuid FK	
uploader_user_id	uuid FK nullable	
uploader_session_id	uuid FK nullable	
type	text	image video
state	text	
source	text	guest professional media_team
storage_key	text	
content_type	text	
byte_size	bigint	
checksum_sha256	text	
width	int	
height	int	
duration_ms	int	
caption	text	
captured_at	timestampz	
variants	jsonb	urls/keys for thumb/md/lg
ai_labels	jsonb	moments/places placeholders
nsfw_score	real	
embargo_until	timestampz	
created_at	timestampz	
deleted_at	timestampz	

Indexes: (event_id, created_at DESC), UNIQUE (event_id, checksum_sha256) WHERE deleted_at IS NULL, (event_id, state)

media_likes

Column	Type	Notes
media_id	uuid FK	
user_id	uuid FK	
created_at	timestampz	
PK(media_id, user_id)		

media_comments

Column	Type	Notes
id	uuid PK	
media_id	uuid FK	
user_id	uuid FK	
body	text	
created_at	timestampz	
deleted_at	timestampz	

media_reports

Column	Type	Notes
id	uuid PK	
media_id	uuid FK	
reporter_user_id	uuid FK	
reason	text	
notes	text	
status	text	open resolved dismissed
created_at	timestampz	

albums

Column	Type	Notes
id	uuid PK	
event_id	uuid FK	
type	text	people places moments professional custom personal
title	text	
cover_media_id	uuid	

Column	Type	Notes
owner_user_id	uuid nullable	personal albums
metadata	jsonb	
created_at	timestamptz	

album_media

Column	Type	Notes
album_id	uuid FK	
media_id	uuid FK	
rank	int	
PK(album_id, media_id)		

face_enrollments

Column	Type	Notes
id	uuid PK	
user_id	uuid FK UNIQUE	one active
consent_version	text	
consented_at	timestamptz	
embedding_ref	text	pointer to secure store / KMS-encrypted blob id
status	text	enrolled revoked
revoked_at	timestamptz	

Never store raw embeddings in plaintext columns.

face_matches

Column	Type	Notes
id	uuid PK	
event_id	uuid FK	
user_id	uuid FK	
media_id	uuid FK	
confidence	real	
suppressed	bool DEFAULT false	user said "not me"
created_at	timestamptz	
UNIQUE(event_id, user_id, media_id)		

resources

Column	Type	Notes
id	uuid PK	
event_id	uuid FK	
title	text	
type	text	pdf link slide other
url_or_media_id	text	
visibility	text	
created_by	uuid	
created_at	timestampz	

certificates

Column	Type	Notes
id	uuid PK	
event_id	uuid FK	
user_id	uuid FK	
title	text	
file_media_id	uuid	
issued_at	timestampz	
UNIQUE(event_id, user_id, title)		

jobs

Column	Type	Notes
id	uuid PK	
event_id	uuid FK	
type	text	highlight export_zip face_backfill purge
state	text	
input	jsonb	
result	jsonb	
error	text	
created_by	uuid	
created_at	timestampz	
updated_at	timestampz	

devices

Column	Type	Notes
id	uuid PK	
user_id	uuid FK	
push_token	text	
platform	text	ios android web
last_seen_at	timestamptz	

notification_prefs

Column	Type	Notes
user_id	uuid FK	
channel	text	push email in_app
category	text	
enabled	bool	
PK(user_id, channel, category)		

notifications

Column	Type	Notes
id	uuid PK	
user_id	uuid FK	
event_id	uuid FK nullable	
category	text	
title	text	
body	text	
data	jsonb	
state	text	
created_at	timestamptz	
read_at	timestamptz	

audit_logs

Column	Type	Notes
id	uuid PK	
actor_user_id	uuid	

Column	Type	Notes
event_id	uuid nullable	
action	text	
target_type	text	
target_id	text	
metadata	jsonb	
created_at	timestampz	

consents

Column	Type	Notes
id	uuid PK	
user_id	uuid FK	
type	text	
version	text	
granted	bool	
granted_at	timestampz	
revoked_at	timestampz	

10.3 Storage (object store)

Not in Postgres: original media bytes, renditions, highlight MP4s, export zips, encrypted embedding blobs.

Keys: `events/{eventId}/media/{mediaId}/original`, `.../thumb.webp`, etc.

10.4 Suggested indexes & partitioning

- Partition `media_objects` by `created_at` or `hash(event_id)` when > 100M rows.
- Partition `face_matches` by `event_id`.
- BRIN on `media_objects.created_at` for large events.
- Cache event landing and wall feed in Redis.

11. Notifications

11.1 Channels

Channel	Use
In-app	Always available when authenticated
Push	iOS/Android/Web push
Email	Transactional + digests
SMS	Invite/RSVP reminders (P1, opt-in)

11.2 Categories & triggers

ID	Category	Trigger	Default channels	Muteable
N-001	event_invite	Invitation sent	email, push	Yes
N-002	rsvp_reminder	T-72h / T-24h before start	email	Yes
N-003	event_live	Event goes live	push, in-app	Yes
N-004	new_media_digest	Burst of new uploads (batched 15m)	push	Yes
N-005	photos_of_you	New face matches ≥ 1	push, email, in-app	Yes
N-006	personal_gallery_ready	Post-event personal delivery	email, push	Yes
N-007	moderation_needed	Report or pending queue > 0	push, in-app	Yes (role)
N-008	highlight_ready	Highlight job succeeded	push, in-app	Yes
N-009	export_ready	Zip/social pack ready	email, push	Yes
N-010	gallery_expiring	T-6h / T-1h before expiry	push, email	No (transactional)
N-011	gallery_expired	Expiry reached	email	No
N-012	role_granted	Became co-org/photographer	email, in-app	Yes

ID	Category	Trigger	Default channels	Muteable
N-013	certificate_issued	Cert available	email, push	Yes
N-014	security	New login / revoke	email	No
N-015	comment_on_your_media	Comment	push, in-app	Yes
N-016	like_milestone	Optional milestones	in-app	Yes

11.3 Templates (copy)

N-005 Photos of you

- **Title:** We found photos of you
- **Body:** {{count}} new photos from {{event_title}}
- **CTA:** View my photos → /e/{{slug}}/me

N-010 Gallery expiring

- **Title:** {{event_title}} closes in {{hours}} hours
- **Body:** Download your favorites before this gallery disappears.
- **CTA:** Open gallery

N-006 Personal gallery ready

- **Title:** Your {{event_title}} photos are ready
- **Body:** We've gathered the memories featuring you.
- **CTA:** Open my gallery

N-001 Invite

- **Subject:** You're invited to {{event_title}} on EventSphere
- **Body:** {{host_name}} invited you to share and collect memories in one place.
- **CTA:** View event

11.4 Batching & quiet hours

Rule	Detail
Digest batching	Collapse N-004 per event per 15 minutes

Rule	Detail
Face match batching	Collapse N-005 to at most 1 / 10 minutes / event while live
Quiet hours	22:00–08:00 local for muteable; N-010/N-014 still send
Deduping	Same <code>dedupe_key</code> not re-sent within TTL

11.5 Preference model

```
{
  "push": { "photos_of_you": true, "new_media_digest": false },
  "email": { "personal_gallery_ready": true, "gallery_expiring": true },
  "in_app": { "*": true }
}
```

Per-event mute: user can mute an event's social notifications without leaving.

11.6 Delivery requirements

Requirement	Target
Push send after trigger	p95 < 30s (non-batched)
Email	p95 < 2 minutes
Failure retry	Exponential backoff, 5 attempts
Unsubscribe	One-click List-Unsubscribe for email marketing/digests
Localization	Templates by <code>user.locale</code>

12. File Handling

12.1 Upload architecture

```
Client → API (init upload) → Signed PUT to Object Storage
      → API (complete) → Processing Queue → Workers
      → variants + AI enqueue → media.state=ready/published
```

Steps

1. Client requests upload session with MIME, size, checksum, idempotency key.
2. API validates quota/type/event state; returns signed URL (TTL 30 minutes).
3. Client uploads directly to object storage.
4. Client calls complete; server verifies object existence + checksum.
5. Worker generates renditions; extracts metadata; optional NSFW scan; AI labeling/face detect.
6. Media becomes viewable per approval mode.

12.2 Supported formats

Kind	Input	Output renditions
Image	JPEG, PNG, WEBP, HEIC	thumb WEBP 320px, md 1280px, lg 2048px, original preserved
Video	MP4, MOV (H.264/HEVC)	poster JPEG, HLS or MP4 720p preview, original
Docs (resources/certs)	PDF, PNG	stored as-is + PDF preview page1 (P1)

12.3 Limits

Constraint	Friends	Celebrations	Professional
Max image	50 MB	50 MB	50 MB
Max video	100 MB	500 MB	500 MB
Max video length	3 min	5 min	5 min
Max concurrent uploads / user	4	8	8
Event total storage soft cap	5 GB	100 GB	2 TB

Soft caps trigger organizer warnings; hard blocks return `UPLOAD_QUOTA`.

12.4 Processing pipeline

Stage	Worker	Failure behavior
Validate magic bytes	sync on complete	reject
Orient / color	image worker	retry
Renditions	image/video worker	retry; gallery may show original if safe
Poster frame	video worker	retry
NSFW score	safety worker	on fail, treat as needs review if approve mode
Object/scene labels	AI worker	nullable
Face detect vectors	AI worker	event-scoped vectors in secure index
Face match	matcher	updates <code>face_matches</code>

12.5 Checksums & dedupe

- Client sends SHA-256; server rehash optional for trusted path.
- Unique (`event_id`, `checksum`) prevents duplicate storage.
- Near-duplicate perceptual hash is P1 and does not block upload; may group in UI.

12.6 Download & signed URLs

Asset	Auth	TTL
thumb/md gallery	member cookie/token or signed cookie	1 hour
original	per <code>download_policy</code>	15 minutes
export zip	owner link	72 hours, single-factor auth
wall CDN	signed path or event wall token	short

Hotlink protection required for originals.

12.7 Retention & purge

1. On event expiry/archive: enqueue `purge` job.
2. Delete object keys + DB rows (or tombstone).
3. Invalidate CDN.
4. Delete event-scoped face index entries.
5. Audit log purge completion.
6. Legal hold skips purge.

12.8 Client responsibilities

- Compress large images before upload when network is cellular (optional setting).
- Respect EXIF orientation.
- Persist upload queue offline.
- Show per-file error states.
- Never trust client-only validation.

12.9 Virus / malware safety

- Block executable MIME/extensions.
- ClamAV or cloud malware scan on docs (P1).
- Strip active content from images where applicable.

12.10 Accessibility of media

- Alt text: optional caption; AI suggested alt P2.
- Video: captions not required v1; poster required.

13. Integrations

13.1 Identity

Provider	Purpose	Priority
Sign in with Apple	iOS/required App Store path	P0
Google OAuth	Web/Android sign-in	P0
Email magic link	Universal fallback	P0
Phone OTP	Optional auth	P1
SAML/OIDC SSO	Corporate Professional	P2

13.2 Cloud infrastructure

System	Purpose
Object storage (S3-compatible)	Media originals & renditions
CDN	Thumbnails, wall assets
Managed Postgres	Primary data store
Redis	Sessions, rate limits, wall fanout
Queue (SQS/PubSub)	Processing jobs
Secrets Manager / KMS	Embedding encryption keys

13.3 AI / ML

Capability	Integration approach
Face detection & embeddings	Managed vision API or in-house model service
Face matching index	Vector DB (event-partitioned)
Scene / moment classification	Vision labeling model
NSFW classification	Safety model
Highlight reel assembly	FFmpeg + ranking service
Story templates	Template renderer

Requirement: AI vendors under DPA; no training on customer media without explicit contract opt-in (default off).

13.4 Communications

Provider type	Use
Transactional email (SES/Postmark/etc.)	Invites, expiry, exports
Push (APNs, FCM)	Mobile push
Web Push	PWA
SMS gateway	RSVP reminders P1

13.5 Calendar

Integration	Priority	Notes
Add to Apple/Google Calendar link	P0	ICS download from event
Google Calendar API sync	P2	Organizer optional

13.6 Social export

Target	Priority	Notes
Save story MP4 / images	P1	Manual share sheet
Instagram / TikTok direct	P2	Where APIs allow
YouTube upload	P2	Highlight reels

13.7 Photographer / DAM (P2)

- Zapier/webhooks for `media.published`
- Lightroom / PhotoShelter import connectors (explore)

13.8 Maps

- Geocoding for event location (Mapbox/Google)
- Place album clustering from EXIF if GPS kept

13.9 Observability

Tool class	Use
APM	API latency
Log drain	Structured logs with <code>request_id</code>

Tool class	Use
Error tracking	Client + server
Product analytics	Funnel: create → share → join → upload → personalize

13.10 Integration security requirements

1. OAuth tokens stored encrypted; minimal scopes.
2. Webhooks signed; retries with backoff.
3. Per-tenant rate limits on inbound webhook storms.
4. No embedding vendor may retain identifiable face images beyond processing SLA (e.g., delete within 24h) unless self-hosted.

14. Security Requirements

14.1 Security goals

Protect event privacy, prevent unauthorized access to media, safeguard biometric face data, and ensure organizers retain control over distribution.

14.2 Authentication

Requirement	Detail
SEC-AUTH-001	Access tokens short-lived (≤ 1 hour); refresh rotatable
SEC-AUTH-002	Magic links single-use, $TTL \leq 15$ minutes
SEC-AUTH-003	OAuth via Apple/Google with nonce/state validation
SEC-AUTH-004	Anonymous tokens scoped to one event; $TTL \leq 24$ hours
SEC-AUTH-005	Logout revokes refresh tokens server-side
SEC-AUTH-006	Step-up auth for account delete and ownership transfer

14.3 Authorization

Requirement	Detail
SEC-AZN-001	Enforce permission matrix server-side on every media URL
SEC-AZN-002	Signed URLs must embed event + media + expiry + user/session
SEC-AZN-003	IDOR tests mandatory for <code>/media/{id}</code> and exports
SEC-AZN-004	Platform admin actions fully audited
SEC-AZN-005	Passcodes stored hashed (Argon2/bcrypt)

14.4 Data protection

Requirement	Detail
SEC-DATA-001	TLS 1.2+ everywhere
SEC-DATA-002	Encryption at rest for DB, object storage, backups
SEC-DATA-003	Face embeddings encrypted with KMS; access tightly limited
SEC-DATA-004	No embeddings in logs, analytics, or client APIs
SEC-DATA-005	PII minimization in logs (hash emails)
SEC-DATA-006	Separate production/staging data; no prod media in staging without anonymization

14.5 Privacy & biometrics

Requirement	Detail
SEC-PRIV-001	Explicit opt-in before any face enrollment
SEC-PRIV-002	Clear retention & revoke UX
SEC-PRIV-003	Cross-event matching off by default
SEC-PRIV-004	Regional compliance: GDPR, UK GDPR, CCPA; biometric policies (e.g., BIPA-aware flows for US)
SEC-PRIV-005	Privacy policy + in-product consent versioning
SEC-PRIV-006	Data subject export & erase endpoints
SEC-PRIV-007	Minors: disable face matching when age_gate=minor

14.6 Application security

Requirement	Detail
SEC-APP-001	OWASP ASVS L2 target
SEC-APP-002	Input validation + output encoding
SEC-APP-003	CSRF protection for cookie sessions (if used)
SEC-APP-004	Strict CSP on web
SEC-APP-005	Rate limiting on auth, join, upload, report
SEC-APP-006	Malware scanning for document uploads
SEC-APP-007	Dependency scanning + SAST in CI
SEC-APP-008	Penetration test before public Celebrations/Pro launch

14.7 Media abuse prevention

Requirement	Detail
SEC-ABUSE-001	Report + moderation workflow
SEC-ABUSE-002	NSFW classifier assist
SEC-ABUSE-003	Ban membership capability
SEC-ABUSE-004	Automatic takedown SLA: critical reports reviewed < 24h

14.8 Secure configuration

Requirement	Detail
SEC-CFG-001	Secrets never in client binaries
SEC-CFG-002	Least-privilege IAM for workers (write only their prefixes)
SEC-CFG-003	Presigned upload URLs constrained by content-type and max size
SEC-CFG-004	Wall kiosk exit passcode

14.9 Incident response

Requirement	Detail
SEC-IR-001	Breach notification runbook
SEC-IR-002	Ability to force-expire all sessions
SEC-IR-003	Ability to disable face matching globally via kill switch
SEC-IR-004	Ability to freeze downloads on a compromised event

14.10 Threat model (summary)

Threat	Mitigation
Guessable event URLs	Unpredictable slugs + optional passcode
Scraping entire galleries	Authz on each asset; rate limits; watermark option
Biometric data leak	Encryption, no API exposure, revoke, vendor DPA
Malicious uploads	Type checks, scan, moderation
Account takeover	Secure auth, session revoke, security emails
Insider access	Audit logs, least privilege, embedding access break-glass

15. Performance Requirements

15.1 Scale targets

Scenario	Target
Friends event	≤ 50 guests, ≤ 2k media
Celebrations	≤ 500 guests, ≤ 10k media
Professional	≤ 10k guests, ≤ 50k media, multi-uploader
Concurrent live wall viewers	5k
Concurrent uploads during peak	500 uploads/minute/event
Platform	Multi-tenant; noisy-neighbor isolation

15.2 Latency SLOs (p95)

Action	SLO
Event landing load	< 1.5s (CDN/TTFB)
Gallery first page (30 items)	< 800ms API + images progressive
Media viewer open	< 300ms to first paint (thumb), full md < 1s on broadband
Join event	< 1s
Upload init	< 400ms
Like/comment	< 300ms
Face match appearance after upload (enrolled user, event ≤10k media)	< 5 minutes p95
Highlight reel 30s (≤500 candidate media)	< 10 minutes
Personal zip (≤500 media)	< 5 minutes
Live wall media appearance after publish	< 10s p95

15.3 Availability

System	Target
API core	99.9% monthly
Uploads	99.5% (degrade gracefully)
Realtime wall	Best-effort; reconnect < 3s

15.4 Client performance

Requirement	Detail
PERF-CL-001	Virtualized gallery lists mandatory above 200 items
PERF-CL-002	Image sizes selected by viewport/DPR
PERF-CL-003	Prefetch next 2 media in viewer
PERF-CL-004	App cold start interactive < 3s on mid-tier phones
PERF-CL-005	Offline queue does not block UI thread

15.5 Backend performance

Requirement	Detail
PERF-BE-001	Cursor pagination only (no large OFFSET)
PERF-BE-002	Hot paths cached: event landing, wall feed
PERF-BE-003	Face index partitioned per event
PERF-BE-004	Thumbnail CDN cache hit ratio > 90%
PERF-BE-005	Async everything heavy (AI, zip, highlights)
PERF-BE-006	DB connection pooling; read replicas for gallery reads

15.6 Load testing gates (pre-launch)

1. 10k media event gallery browse simulation.
2. 500 concurrent uploaders for 10 minutes.
3. Wall fanout to 5k SSE/WS clients.
4. Face backfill on 50k media for 2k enrolled users.

15.7 Degraded modes

Condition	Mode
AI down	Gallery works; personalization delayed banner
Storage elevated errors	Read-only media; pause new uploads with message
Queue backlog	Show processing states; no user data loss

16. Acceptance Criteria

Format: **AC--**. Must pass for the related feature to ship.

16.1 Auth & account

ID	Criteria
AC-AUTH-01	User can sign in with Apple, Google, or magic link and land on My Events.
AC-AUTH-02	Invalid/expired magic link shows clear error and allows resend.
AC-AUTH-03	Logout invalidates session on server; subsequent API calls return 401.
AC-AUTH-04	Account deletion blocked with <code>OWNERSHIP_BLOCK</code> if user still owns events.
AC-AUTH-05	After account delete, user cannot sign in; face embeddings removed within 24h.

16.2 Event create & join

ID	Criteria
AC-EVT-01	Organizer creates Friends event in ≤ 8 taps/clicks with 48h retention prefilled.
AC-EVT-02	System generates unique slug, join URL, and QR.
AC-EVT-03	Guest joins via QR without installing an app (PWA/web).
AC-EVT-04	Wrong passcode fails with <code>PASSCODE_INVALID</code> and does not reveal event media.
AC-EVT-05	Banned user cannot rejoin.
AC-EVT-06	Use case cannot change after first guest join.

16.3 Upload & gallery

ID	Criteria
AC-MED-01	Guest can upload 10 photos in one batch with visible per-item progress.
AC-MED-02	Unsupported file type rejected with <code>VAL_MEDIA_TYPE</code> before/at complete.
AC-MED-03	Duplicate checksum in same event returns <code>DUPLICATE_MEDIA</code> .
AC-MED-04	In auto-approve mode, media appears in gallery after processing without moderator action.
AC-MED-05	In approve-each mode, other guests cannot see media until approved.
AC-MED-06	Uploader can delete own media; it disappears from gallery and wall within 60s.
AC-MED-07	Gallery supports $\geq 5,000$ items without UI freeze (virtualization).
AC-MED-08	Offline-selected uploads resume when connectivity returns (idempotent).

16.4 Social

ID	Criteria
AC-SOC-01	Like toggles and updates count for other clients within 10s (or on refresh).
AC-SOC-02	Comment max length enforced; empty comment rejected.
AC-SOC-03	Report creates moderation queue item for organizers.

16.5 Personalization

ID	Criteria
AC-FACE-01	Without consent, enrollment cannot proceed.
AC-FACE-02	After enrollment, Photos of You shows matches with confidence above threshold.
AC-FACE-03	"Not me" removes media from personal gallery and suppresses rematch.
AC-FACE-04	Revoke consent deletes enrollment; personal gallery empties; matching stops.
AC-FACE-05	Anonymous users are prompted to create an account before enrollment.
AC-FACE-06	In a fixture with 5,000 photos and 2 true matches, personal gallery returns those 2 (allowing ≤ 1 false positive before feedback).

16.6 Friends expiry

ID	Criteria
AC-FR-01	Countdown visible on event overview.
AC-FR-02	N-010 sends at T-6h.
AC-FR-03	After expiry, uploads return <code>EVENT_EXPIRED</code> .
AC-FR-04	After purge job, media URLs return 404/410 and storage objects are gone.

16.7 Celebrations

ID	Criteria
AC-CEL-01	Guest can RSVP going/maybe/declined; organizer sees updated list.
AC-CEL-02	Live gallery updates for a second client within 10s of publish.
AC-CEL-03	Photo wall shows only published media.

ID	Criteria
AC-CEL-04	Branding theme applies to join landing and hub header.

16.8 Professional

ID	Criteria
AC-PRO-01	Photographer album can be embargoed then published atomically.
AC-PRO-02	Admin face search returns media for a target enrolled attendee.
AC-PRO-03	Certificate issued to user appears in My Certificates and downloads via signed URL.
AC-PRO-04	Personal gallery ready email deep-links to Photos of You.
AC-PRO-05	Resources hub lists only visible resources for guests.

16.9 Roles

ID	Criteria
AC-ROLE-01	Co-organizer can moderate but cannot hard-delete event.
AC-ROLE-02	Guest cannot open analytics (403).
AC-ROLE-03	Role changes audited in <code>audit_logs</code> .

16.10 Security & privacy

ID	Criteria
AC-SEC-01	Direct media URL without auth/signature fails.
AC-SEC-02	User A cannot access User B's personal gallery.
AC-SEC-03	Embeddings never appear in any API JSON.
AC-SEC-04	Rate limit returns 429 with friendly message.

16.11 Performance gates

ID	Criteria
AC-PERF-01	Gallery first page p95 < 800ms in staging load test (warm cache).
AC-PERF-02	Wall update p95 < 10s under 1k concurrent viewers.
AC-PERF-03	App remains usable when AI worker is stopped (no hard dependency).

16.12 Accessibility (baseline)

ID	Criteria
AC-A11Y-01	Critical flows usable with VoiceOver/TalkBack (join, upload, gallery).
AC-A11Y-02	Focus order logical on web join + create wizard.
AC-A11Y-03	Text contrast meets WCAG AA for primary UI.

17. Wireframes / Mockups

Structural wireframes for product and design. Layout, hierarchy, and components only — not final visual design.

Design system should keep brand-first event landings and avoid generic purple-gradient AI aesthetics.

17.1 S-AUTH-01 Welcome (mobile)

```
+-----+
|           |
|  EventSphere  |  <- brand hero signal
|           |
| Every event.  |
| One digital home. |
|           |
| [ Get started ] |
| [ Sign in ]   |
|           |
| Or join with QR |
| [ Scan code ]  |
|           |
+-----+
```

17.2 S-HOME-01 My Events

```
+-----+
| EventSphere      (bell) |
| Hello, Alex      |
|           |
| [ Live ] [ Upcoming ] [ Past ] |
|           |
| +-----+ |
| | cover      | |
| | Amritsar Trip | |
| | Expires in 14h | |
| | 5 people · 48 photos | |
| +-----+ |
| +-----+ |
| | Sara & Dan Wedding | |
| | Sat · 120 going   | |
| +-----+ |
|           |
| [ + Create ] [ Join ] |
|           |
+-----+
| Home   Create   Join   Me |
+-----+
```

17.3 S-CRT-01 Choose use case

```
+-----+
| Create an event |
| What are you planning? |
| |
| +-----+ |
| | Friends | |
| | Trips, dinners, | |
| | small gatherings | |
| +-----+ |
| +-----+ |
| | Celebrations | |
| | Birthdays, weddings | |
| +-----+ |
| +-----+ |
| | Professional Events | |
| | Conferences, corp... | |
| +-----+ |
+-----+
```

17.4 S-JOIN-01 Event landing

```
+-----+
| ##### cover image ##### | <- full-bleed
| ##### |
| |
| Amritsar Trip |
| Hosted by Alex |
| Fri-Sun · Private |
| |
| Share memories in one |
| place – no AirDrop chaos. |
| |
| [ Join event ] |
| Already have an account? |
| Sign in |
+-----+
```


17.5 S-EVT-01 Event overview

```
+-----+
| <- Amritsar Trip      share |
| Gallery closes in 14:02 |
|
| [ Upload photos ] |
| [ Find photos of me ] |
|
| Recent
| [ ][ ][ ][ ][ ][ ][ ][ ][ ] |
|
| Albums
| People · Places · Moments |
+-----+
| Home  Gallery  Upload More|
+-----+
```

17.6 S-EVT-02 Gallery grid

```
+-----+
| Gallery              filter |
| All | People | You | Pro |
|
| [ ][ ][ ][ ][ ][ ][ ][ ][ ] |
| [ ][ ][ ][ ][ ][ ][ ][ ][ ] |
| [ ][ ][ ][ ][ ][ ][ ][ ][ ] |
|
| (virtualized scroll) |
+-----+
```

17.7 S-EVT-05 Photos of you

```
+-----+
| Photos of you
| 12 photos · Updated now
|
| [ Download all ]
|
| [ ][ ][ ][ ][ ][ ][ ][ ][ ] |
|
| Tip: Tap a photo ->
| "Not me" if wrong
+-----+
```

Not enrolled empty:

Photos of you Automatically find pictures that include you. [Enable photo matching] [Not now]
--

17.8 S-EVT-06 Media viewer

X ... [full media] <3 24 Chat 3 Down Uploaded by Sam · 7:42pm
--

17.9 S-EVT-08 Upload composer

Upload Done [Camera] [Library] Selected (12) [][][][][][][] [][][][][][][] Optional caption [Upload 12 items]
--

17.10 S-EVT-09 Upload progress

```
+-----+
| Uploading 8 of 12...
| [#####----] 67%
|
| IMG_01.jpg      Done
| IMG_02.heic    45%
| VID_03.mov      Waiting
| IMG_04.jpg      Failed R
|
| Keep the app open for
| large videos.
+-----+
```

17.11 S-EVT-10 Guests & RSVP (Celebrations)

```
+-----+
| Guests
| Going 120 · Maybe 14
|
| [ Invite ] [ QR ]
|
| Search guests...
|
| Ava Patel      Going
| Ben Okonkwo    Maybe
| Chris Lee      --
+-----+
```

17.12 S-EVT-22 Share & QR

```
+-----+
| Share event
|
| +-----+
| | QR |
| +-----+
|
| eventsphere.app/e/...
| [ Copy link ]
| [ Download QR ]
| [ Print table cards ]
| [ Share... ]
+-----+
```

17.13 S-WALL-01 Live photo wall (kiosk)

```
+-----+
| EventSphere · Sara & Dan |
|                               |
|  #### img ####   #### img #### |
|  #####          |#####          |
|                               |
|          ##### new photo ##### |
|                               |
|  #### img ####   #### img #### |
|                               |
+-----+
(no cards, no badges over media; ambient motion only)
```

17.14 S-EVT-13 Highlights

```
+-----+
| Highlights |
| [ Generate recap ] |
| Ready |
| > Weekend Recap 0:32 |
| [ Export story ] [ Share ] |
| Processing... |
| [###----] highlight |
+-----+
```

17.15 S-EVT-16 Certificates (Professional)

```
+-----+
| Certificates |
|             |
| +-----+ |
| | Certificate of | |
| | Attendance | |
| | [ Preview ] [ Download] |
| +-----+ |
|             |
| Resources |
| Slides.pdf |
| Speaker notes (link) |
+-----+
```

17.16 S-EVT-19 Event settings

Event settings	
Basics	>
Access & privacy	>
Moderation	>
Retention & expiry	>
Branding	>
Roles	>
Live wall	>
Analytics	>
Danger zone	>

17.17 S-PROF-03 Privacy & face data

Privacy & face data	
Photo matching	ON
Find photos that include you.	
Cross-event matching	OFF
[Manage enrollment]	
[Delete my face data]	
Learn how this works	>

17.18 Desktop Event Hub (Celebrations / Pro)

EventSphere		Sara & Dan Wedding				Share	QR
Overview	Live · 482 guests · 1,902 photos						
Gallery							
Guests	[Upload] [Photos of you] [Wall]						
Highlights							
Resources	[] [] [] [] [] [] [] [] [] [] [] []						
Settings	[] [] [] [] [] [] [] [] [] [] [] []						

17.19 Navigation diagram

```
flowchart LR
  Welcome --> SignIn
  Welcome --> JoinQR
  SignIn --> Home
  Home --> Create
  Home --> EventHub
  JoinQR --> EventLanding
  EventLanding --> EventHub
  Create --> EventHub
  EventHub --> Gallery
  EventHub --> Upload
  EventHub --> PhotosOfYou
  EventHub --> Guests
  EventHub --> Settings
  Gallery --> MediaViewer
  PhotosOfYou --> MediaViewer
  Settings --> ShareQR
  Settings --> Wall
```

17.20 Design notes for mockup phase

1. **Brand first** on Welcome and Event Landing — product/event name is the hero signal.
2. **Full-bleed** cover on join landing; no inset hero cards.
3. **No cards in hero**; gallery grids are media themselves, not card chrome.
4. **One job per screen**: Upload composer does not also show analytics.
5. Motion: subtle wall crossfade; gallery like micro-interaction; expiry countdown used sparingly.
6. Avoid default purple gradients, cream+terracotta cliché, and broadsheet dense layouts.

18. Glossary of Terms

Term	Definition
EventSphere	The product — an AI-powered digital home for event memories.
Event	A time-bound container for people, media, and related digital assets.
Use case	Product framing for capability bundles: Friends, Celebrations, Professional Events (not pricing tiers in narrative).
Friends	Small-group use case with temporary shared galleries (default 48h expiry).
Celebrations	Mid-size hosted events: weddings, birthdays, university events; QR, RSVP, live wall, branding.
Professional Events	Large/org events: conferences, corporate events, festivals; face search at scale, certs, media hub.
Organizer / Host	User who owns the event and has full event admin rights.
Co-Organizer	Privileged helper; cannot hard-delete or transfer ownership.
Guest / Attendee	Participant with view/upload rights per event policy.
Anonymous Guest	Joined without account via short-lived session token.
Photographer	Role for professional album uploads and delivery.
Media Team	Role for highlights, promo assets, and hub content.
Moderator	Role limited to content review/actions.
Membership	Relationship between a user/session and an event, including role and status.
Gallery	Primary browsable collection of event media.
Album	Subset of media grouped by people, places, moments, professional source, or custom.
Photos of You / Personalized gallery	Private album of media matched to an enrolled user's face.
Face enrollment	Opt-in capture of selfies used to create an encrypted face embedding.
Face embedding	Mathematical representation of a face used for matching; never exposed via API.
Face matching	Process of linking media to enrolled users by similarity.
Live photo wall	Large-display / kiosk surface showing approved media in near real time.
Highlight reel	Auto-generated short video recap of event moments.
Story template	Vertical export format optimized for social stories.
QR join	Entering an event by scanning a QR code that opens the join landing.
Passcode gate	Optional shared code required before join.
RSVP	Invitation response: going, maybe, declined.

Term	Definition
Approval mode	<code>auto</code> publishes uploads immediately; <code>approve_each</code> requires moderator approval.
Embargo	Professional media hidden until organizer publish time.
Retention	How long media remains available before expiry/purge.
Expiry	State when retention elapses; uploads stop; purge scheduled.
Purge	Irreversible deletion of media bytes and related derived data.
Legal hold	Flag preventing purge for compliance/investigation.
Resource	Non-gallery digital asset (PDF, link, slides) attached to an event.
Certificate	Per-attendee issued document claimable after/during a Professional event.
Digital asset hub	Professional unified area for photos, videos, highlights, certs, resources, brand kit.
Brand kit / branding	Event theme: colors, logo, cover, approved fonts.
Export pack	Generated zip or social pack of selected media.
Signed URL	Time-limited authorized URL for direct storage access.
Idempotency key	Client key ensuring retries do not create duplicate uploads/jobs.
Moderation queue	List of reported or pending media awaiting organizer action.
NSFW score	Safety model estimate that media may be inappropriate.
Deep link	URL that opens a specific event/screen in app or web.
Slug	Human-readable unique event path segment (<code>/e/{slug}</code>).
Viewer membership	The current user's role/status on an event, returned in API payloads.
Cross-event matching	Optional consent to reuse embeddings across multiple events (default off).
Tombstone	Minimal metadata retained after purge for support/audit for a limited period.
Platform Admin	Internal EventSphere operator with audited break-glass tools.
Organization	Multi-event owning entity for companies/agencies (schema-ready).
P0 / P1 / P2	Priority: must-have MVP, v1, post-v1.

Pitch line (canonical)

We don't replace your camera. We don't replace your photographer. We don't replace Instagram.

We give every event one digital home.

19. Assumptions & Open Questions

Captured so engineering/design can proceed without blocking; resolve before build lock.

Assumptions (used throughout this PRD)

1. v1 ships as **mobile-first PWA + web**, with native iOS/Android as P1.
2. **Use cases** (Friends / Celebrations / Professional) are capability templates, not customer-facing price names.
3. Friends default retention is **48 hours** after event end.
4. Face matching is **opt-in** and never required for core gallery use.
5. Direct-to-cloud uploads with signed URLs are the upload path.
6. Postgres + object storage + queue workers are the reference architecture.
7. English-first UI; i18n hooks from day one.

Open questions

ID	Question	Impact
OQ-01	Exact monetization model later — subscriptions, per-event, storage overage?	Billing schema
OQ-02	Friends: can any member extend expiry, or organizer only?	BR-EVT / UF-08
OQ-03	After Friends expiry, is there a 24h read-only grace or immediate lock?	State machine
OQ-04	Target regions for launch (biometric law variance)?	Consent flows
OQ-05	Self-host face models vs vendor API?	Cost, privacy, latency
OQ-06	Max video length for Celebrations/Pro — 5 min enough?	Upload UX
OQ-07	Will ticketed events need Eventbrite/integration in v1?	Currently out of scope
OQ-08	Watermarking default on/off for Professional guest downloads?	Export policy
OQ-09	Anonymous upload claiming across devices without account — support?	Identity model
OQ-10	Highlight music licensing approach?	Reel feature

Decision log

Date	Decision
2026-07-25	Initial PRD draft created from EventSphere pitch
2026-07-25	Do not pitch Free/Paid/Enterprise; use Friends / Celebrations / Professional Events